PROMPTS Which I Used To Build This Project

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Stage 1: Prompt:

Build a Python Streamlit app that scrapes Australian medical clinics from a URL, lists them in a dashboard, and uses an LLM to generate cold outreach emails to practice managers.

Issues Occurred During Stage 1

Category	Issue Encountered	Real-World Failure Impact
AI Tone & Hallucination	Generic US-style marketing jargon: The model used terms like "physicians", "pediatricians", "hospital networks", and American spellings (center, utilize).	Instantly recognized as spam and deleted by Australian Practice Managers and Principal GPs.
Medicare / MBS Blindness	No understanding of Australian healthcare economics: Failed to recognize Medicare billing structures (Bulk Billing vs. Mixed Billing vs. Private Billing).	Suggested solutions that directly clashed with the clinic's business model (e.g., pitching expensive private billing tools to a 100% bulk billing practice).
Scraper Instability	Regex & DOM crashes: Standard US phone regexes failed on Australian phone structures (07 xxxx xxxx, +61, 1300). Network drops crashed the script.	App crashed with unhandled HTTP/DOM exceptions when a clinic URL was unreachable or dynamic.
Hard API Key Dependency	No fallback logic: If OPENAI_API_KEY was missing, the entire application threw unhandled API authentication exceptions.	Inability to evaluate, demo, or run the app offline or in sandboxed review environments.
UI Contrast & Streamlit Bugs	BaseWeb white-on-white text collision: Streamlit default styling conflicted between dark/light elements. Dropdowns and inputs became illegible.	Unusable UI with invisible form inputs and lack of clinical data hierarchy.

Stage 2: Prompt:

Refactor ClinicIQ into modular files: app.py, ai_engine.py, scraper.py, and data/seed_clinics.py.

Add Australian phone regex (07, 02, 1300, +61) and postcode parsing.

Include pre-seeded Australian GP clinics from Queensland (Robina, Gold Coast, Brisbane).

Add error handling to the scraper and allow filtering by State and Billing Policy in Streamlit.

What Was Solved from Stage 1

- Modular Codebase: Clean separation of concerns between DOM parsing (scraper.py), LLM orchestration (ai_engine.py), and dashboard presentation (app.py).

- Australian Geography & Phone Parsing: Accurate Australian regex matching for landlines and postcodes (QLD 4226, +61 7 5689 1234).
- Seed Data: Curated dataset (INITIAL_CLINICS) allowing immediate testing without manual web scraping.
- Basic Filtering: State and Billing Policy dropdowns added to the Practice Directory.

Issues & Gaps Still Remaining in Stage 2

Category	Remaining Bottleneck	Why It Needed Refinement
One-Size-Fits-All Outreach	Lack of Campaign Segmentation: The prompt pitched both doctors and software simultaneously, creating a confusing, diluted value proposition.	Practice Managers reject pitches that lack a clear, singular operational focus.
Uncalibrated Lead Scoring	Subjective evaluation: No quantified lead scoring (e.g. 50–99) or structured operational gap analysis (e.g. weekend doctor burnout vs. receptionist phone queues).	Clinics could not be prioritized for sales outreach in CRM pipelines.
Output Parsing Failures	Unstructured plain text: Output had varying layouts, missing dedicated subject lines, and lacked follow-up cadences.	Inability to cleanly export structured payloads into CRM software or JSON files.
Offline Fragility	Incomplete Mock Fallback: When network requests were blocked or OpenAI quotas ran out, the output was empty rather than providing heuristic data.	The app required 100% live uptime and active credits to demonstrate capabilities.
Streamlit Popover CSS Bugs	Dropdown menus still turning white-on-white: BaseWeb popovers rendered outside .stApp DOM tree retained unreadable styles.	Assessors and users on different OS/browser color schemes experienced invisible menu items.

(Stage 3): Production-Grade, RACGP-Calibrated Final System

1. What Was Solved from Stage 2

1. RACGP & MBS Domain Calibration: Integrated Australian General Practice standards (Vocationally Registered/VR GPs, Medicare rebate freezes, gap fees, Best Practice / MedicalDirector software familiarity, Australian spelling).

2. 3 Distinct Campaign Tracks:

- Track 1: GP Recruitment & Locum Roster Coverage (solves weekend 7-day doctor shortages).
- Track 2: AI Front-Desk & Triage Automation (solves peak receptionist phone queues and patient intake).
- Track 3: Allied Health & Practice Expansion (expands clinical throughput and mixed billing revenue).

3. Structured JSON Output Schema: Enforced JSON schema for lead scoring (50–99), operational gap identification, and distinct SUBJECT:, BODY:, and FOLLOW_UP_NOTE: markers for CRM automation.

4. Zero-Crash Dual Engine (Live AI + Heuristic Fallback): If OpenAI API key is unavailable or the sandbox blocks network traffic, an intelligent rule-based engine activates automatically with Australian clinical reasoning.

5. Robust Scraper Heuristics: Heuristic detection of 7-day operating hours, Bulk/Mixed/Private billing classifications, and domain fallback simulators.

6. Pixel-Perfect Custom UI Theme: Injected deep CSS overrides across .stApp, [data-baseweb="select"], and popover listboxes for guaranteed contrast across all devices.

2. Stage 3 Final Internal AI Engine Prompts (As Implemented in ai_engine.py)

System Prompt

You are an elite Australian Healthcare Growth Strategist and Practice Management Consultant with 15+ years experience in the Australian General Practice (RACGP) ecosystem.

You understand:

1. Australian Medicare (MBS billing items, Bulk Billing vs Mixed Billing pressures, gap fees).

2. GP staffing shortages, AHPRA compliance, VR (Vocationally Registered) vs Non-VR GPs, and after-hours medical services.

3. Clinic operations: Practice Managers, Principal Doctors, and front-desk phone congestion.

Your goal is to analyze Australian clinic profiles, identify operational bottlenecks, and craft compelling, ethical, professional B2B outreach that converts without sounding like spam. Always write in professional Australian business English (e.g. utilise, prioritise, centre).

Outreach Generator Prompt Template

Draft a high-converting, professional B2B email to the Practice Manager or Principal GP at {clinic_name}.

OUTREACH TYPE: {campaign_type}

Options: "GP Recruitment & Locum Coverage" | "AI Front-Desk & Triage Automation" | "Allied Health Partnership"

CLINIC CONTEXT:

- Clinic: {clinic_name}

- Suburb: {suburb}, {state}

- Billing: {billing_type}

- Operating Hours: {operating_hours}

- Primary Bottleneck / Opportunity: {growth_opportunity}

CONSTRAINTS & GUIDELINES:

1. Tone: Respectful, consultative Australian medical executive tone. Never pushy or salesy.

2. Personalization: Explicitly reference their specific hours (e.g., weekend coverage if open 7 days), location in {suburb}, and billing format.

3. Structure:

- Compelling, low-friction Subject Line (under 9 words, e.g. "Quick question regarding weekend GP coverage in {suburb}").
- Salutation to "Dear Practice Manager & Clinical Director".
- Specific observation about their practice.
- Clear value proposition with measurable outcome (e.g. reducing wait times, staffing after-hours, increasing billings).
- Frictionless Call-to-Action: Suggest a 10-minute intro Zoom or a brief reply.

4. Do NOT make unrealistic medical claims. Respect RACGP guidelines.

Return the email formatted with:

SUBJECT: <subject line>

BODY:

<email body>

FOLLOW_UP_NOTE: <a brief 1-sentence follow-up message to send 3 days later if no reply>

3. Stage 3 Master Project Prompt (To Re-Generate or Expand ClinicIQ End-to-End)

If you need a single comprehensive master prompt to scaffold or instruct an AI agent to build this entire application from scratch, use the following production specification:

You are a senior full-stack AI engineer and healthcare product specialist.

Build a production-grade Python web application named "ClinicIQ" (Australian General Practice Intelligence & AI Growth Automator) using Streamlit, BeautifulSoup4, and OpenAI.

PROJECT SPECIFICATION & ARCHITECTURE

1. DOMAIN & TARGET MARKET

Tailored for the Australian healthcare market (RACGP guidelines, Medicare MBS, Bulk Billing vs Mixed Billing, AHPRA compliance, VR GPs).

Focus on General Practices in Queensland (Robina, Gold Coast, Brisbane, Southport) and national primary care networks.

2. SYSTEM COMPONENTS & FILE STRUCTURE

app.py: Multi-tab Streamlit dashboard (Practice Directory, Data Ingestion, Outreach Engine, Prompt Lab, Practice Analytics) with custom CSS ensuring 100% readable, high-contrast dark text (#14202E) on white inputs (#FFFFFF) with Deep Navy (#0F1F2E) and Teal (#0E7C7B) brand accents.

ai_engine.py: Dual-mode LLM engine (OpenAI GPT-4o-mini + Offline Heuristic Fallback) supporting:

- Lead scoring (50-99) and clinical gap detection.
- b) Multi-angle B2B outreach generation (GP Recruitment, AI Front-Desk Triage, Allied Health Expansion).
- c) Prompt Registry exposing System Prompts and Version History (V1 vs V2 vs V3).

scraper.py: Web extraction module parsing Australian clinic websites for titles, Australian phone numbers (07, 02, 1300, +61), states, postcodes, operating schedules, and billing models with resilient fallback handling.

data/seed_clinics.py: Pre-loaded verified Australian GP clinic dataset (e.g. Robina 7 Day Doctors, Gold Coast Family Medical, Burleigh Waters Health).

3. KEY WORKFLOWS

Directory: Filter practices by State and Billing model, view operating hours, doctors on staff, and run live AI lead scoring.

Scraper: Enter any clinic URL; extract structured metadata with live preview and "Add to Database" functionality.

Outreach Engine: Select clinic and campaign objective, generate personalized Subject, Body, and Day-3 Follow-Up, with one-click JSON CRM export.

Prompt Lab: Interactive inspector showing system prompts, temperature settings (0.3-0.4), and iterative prompt failure modes.

Analytics: Distribution charts for billing models and states with CSV dataset download.

4. RELIABILITY REQUIREMENTS

The system MUST run completely error-free even when no OpenAI API key is provided, using contextual fallback heuristics that mimic LLM outputs.

Scraper must never crash on network failure, returning heuristic fallback structures instead.